

Correct Box, Invented Steps

DATA 443/643 · Week 13, note 13.3

Week 13

American University

Ahmad Mousavi

Ideas from CS224N 2026 L12–L13 (unfaithful CoT; process vs outcome). Original slides; that course is not copied.

Lecture plan

- 1 Process vs outcome
- 2 Chen et al.: CoT need not match the hint used
- 3 52 in the steps, 42 in the box
- 4 Checker pass rate among wins

Key goal: accuracy +1 can still be faithfulness 0.

First time: what / why / formula

- What / why. Lucky win: last step says 52, box says 42, gold is 42. Accuracy can pass while the trace is a lie. Process vs outcome (Lightman et al.). CoT and SC can look careful and still be unfaithful. A project that “shows work” must check the work.
- Architecture. No new net. A checker: parse intermediates vs the box. Optional process reward (score steps), vs outcome-only.
- Formula. Cartoon: $F = \#\{\text{wins with consistent trace}\} / \#\{\text{wins}\}$. Accuracy alone is not F .
- Tradeoffs. + Catches theater. – Checkers can be wrong; humans are slow; process rewards are extra training (not this lab).

How. Lab 13 flags the 52→42 row. Report accuracy and faithful-among-wins.

Two different claims

- Outcome: the box matches gold.
- Process: the written steps compute that box, validly.
- Lightman et al.: a process reward scores steps, not only the box.

Faithfulness: do the steps actually cause the answer?

trace looks careful
(but the algebra is wrong)

final number happens
to be correct

Faithfulness among correct boxes

$$F = \frac{n_m}{n_w}$$

n_w : right box. n_m : steps also match the box.

Not a claim about inner states. We do not read residuals.

A majority of 42 can still be three lucky wins.

Lab 13's lucky win

- Steps: $23 + 10 = 33$, $33 + 9 = 52$.
Box: 42. Gold: 42.
- Last integer in steps is 52. Unfaithful.
Lucky win.
- Chen et al. 2025: models can use a hint they never write.

Correct box, wrong last step: accuracy +1, faithfulness 0.

field	value
last integer in steps	52
box / answer	42
gold	42

What the number is not

Answer accuracy

Did the box match gold? Lucky 42 still scores.

Checker pass rate

Among those wins, did the steps match the box? Put that in the report.

What to take from this lecture

Parse

Last integer in the steps vs the answer field.

Product

Do not ship an unfaithful trace as “showing work.”

Lab 13

Flag the 52→42 row. Fluency is not the check.